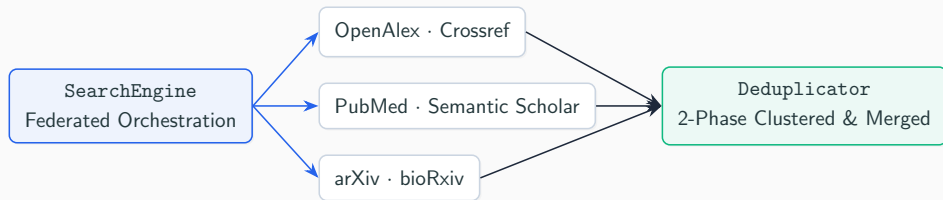


Building a Reproducible Scholarly Search Toolkit

Episode 15: The Federated SearchEngine

Nexus Scholar Suite

Federated Search Architecture



Federation Snowballing API

1. Multi-Provider Discovery

```
engine = SearchEngine(providers=["openalex", "crossref", "arxiv"])
query = Query("quantum error correction", max_results=50)
results = engine.search(query) # Deduplicated Document list
```

2. Citation Snowballing

```
citing_works = engine.snowball(
    paper_id="W2741809807", direction="forward", provider="openalex"
)
```